

FACTSHEET FS10 · Betting against beta

Lever up the low-beta stocks, short the high-beta ones

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STRATEGY	ORIGIN	FAMILY
Defensive / betting against beta	Frazzini & Pedersen (2014)	Low risk
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (low-beta stocks)
REBALANCE	STATUS	
Monthly	Descriptive factsheet, not a registered trial	

The claim. Low-beta stocks have higher risk-adjusted returns than high-beta stocks. A portfolio that is long leveraged low-beta stocks and short de-leveraged high-beta stocks, beta-neutral, earned a large positive return in the US 1926–2012 and in 19 other markets (Frazzini & Pedersen 2014). **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned +2.0% to +9.5% a year and was positive in all six; it passes the 2.87 gate in UK. The long-only book (low-beta stocks) had a higher Sharpe than the equal-weight universe in 5 of the six (EU, UK, DK, SCANDI, World), with alphas of +1.6% to +3.8%, passing the gate in UK. **What goes wrong (§10):** the main drags are a weaker second half (2020–26). None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Low risk; its returns sit closest to the low risk cluster of the signal library. **360° view (§12):** nearest neighbours Volatility (252 days), Volatility; after its closest neighbours and the market it keeps an alpha of -9.6% to +3.9% (largest |t| 3.6), so it carries some information of its own.

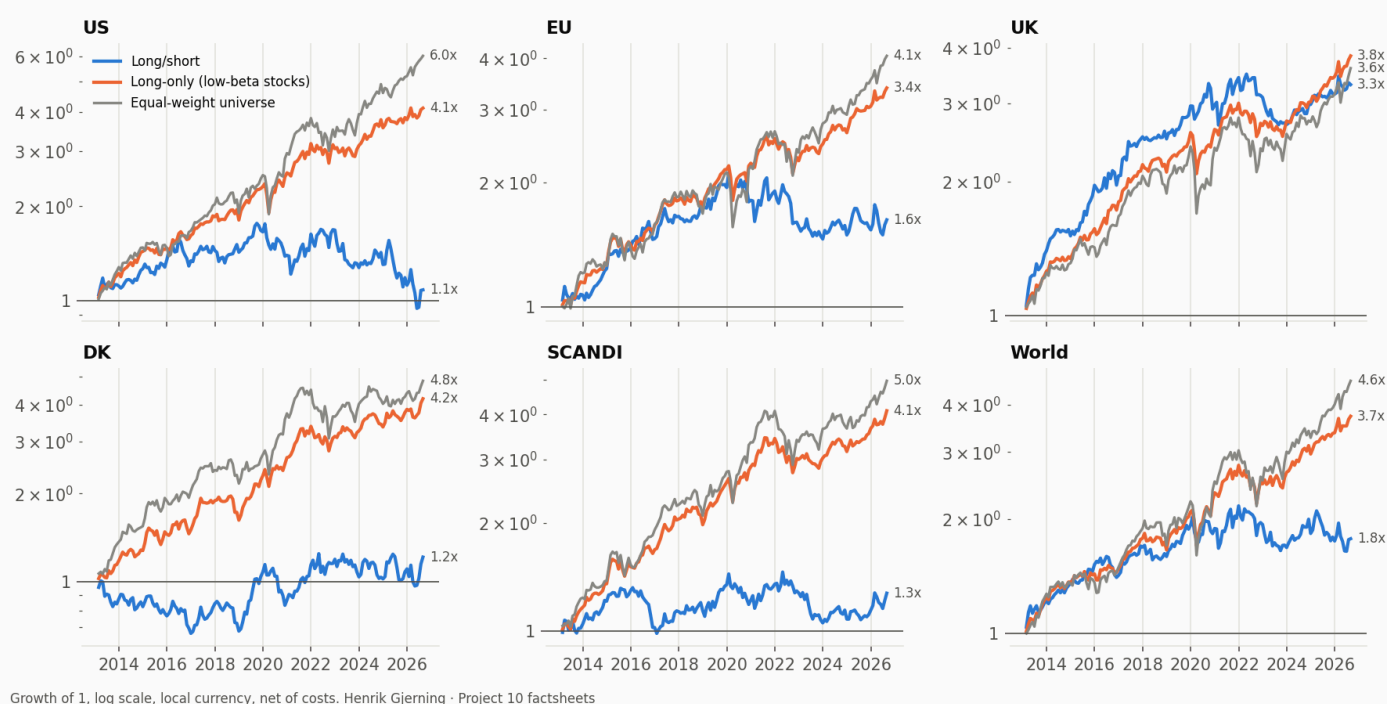
1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
US	501 / 10	+2.0%	0.54	-0.8% (-0.2)	0.19	+11.0%	0.95	-19%	+14.1%	0.95	+15.3%
EU	238 / 10	+4.4%	1.55	+3.6% (1.2)	0.07	+9.4%	0.93	-19%	+10.9%	0.78	+10.1%
UK	239 / 10	+9.5%	3.16	+7.9% (2.7)	0.16	+10.4%	1.07	-19%	+9.9%	0.74	+8.5%
DK	19 / 3	+2.9%	0.66	+2.6% (0.6)	0.02	+11.1%	0.91	-18%	+12.3%	0.82	+10.3%
SCANDI	62 / 5	+2.5%	0.87	+1.9% (0.6)	0.05	+10.9%	1.02	-20%	+12.5%	0.91	+10.7%

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
World	971 / 10	+5.1%	1.84	+1.9% (0.6)	0.26	+10.2%	0.90	-21%	+11.9%	0.80	+10.8%

Monthly, local currency (World in USD). L/S = equal-weighted low-beta stocks minus high-beta stocks (rank-weighted, each leg scaled to beta 1), net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Betting against beta on six stock universes, 2013-2026



2. Strategy description

Low-beta stocks have higher risk-adjusted returns than high-beta stocks. A portfolio that is long leveraged low-beta stocks and short de-leveraged high-beta stocks, beta-neutral, earned a large positive return in the US 1926–2012 and in 19 other markets (Frazzini & Pedersen 2014).

Why it might work. Leverage-constrained investors buy high-beta stocks instead of levering the market, bidding them up (Black 1972; Frazzini & Pedersen 2014); lottery demand (Bali, Brown, Murray & Tang 2017).

Rule	Original (Frazzini & Pedersen (2014))	This factsheet
Signal	Shrunk beta from 1-year volatility and 5-year correlation	252-day beta vs the equal-weight universe (no shrinkage)
Portfolio	Rank-weighted legs above/below median beta, each scaled to beta 1	Same construction
Weighting	Rank weights	Rank weights

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP <code>closeadj</code> (total return), 805 tickers incl. delisted	Point-in-time: top 500 by market cap each month-end (Sharadar filings × price, A06 method)	~499	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** Financing uses the USD risk-free rate for all markets, which overstates the cost in EU and Denmark while their rates were negative (2015–21). Betas are raw, not shrunk.

4. Signal creation

$BAB_{t+1} = r_L/\beta_L - r_H/\beta_H$, with rank-weighted low- and high-beta legs (the net long cash is financed at the USD risk-free rate + 0.5% a year; the high-beta leg pays a size-tiered borrow fee). Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).

- **Long/short** = low-beta stocks minus high-beta stocks; **long-only** = low-beta stocks; benchmarks: equal-weight universe and size-weighted universe.
- Frazzini & Pedersen construction: betas ranked, each leg weighted by rank distance from the median and scaled to an ex-ante beta of 1; no risk-free rate is subtracted.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (top 500, point-in-time), median 501 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.0%	+0.6%	16.6%	0.12 [-0.32, 0.58]	0.05	-46%	52%	0.54	0.19	-0.8% (-0.2)	9.9%
Long-only (net)	+11.1%	+11.0%	11.8%	0.95 [0.55, 1.43]	1.51	-19%	64%	5.38	0.66	+1.6% (1.1)	7.0%
Low-beta stocks (gross)	+11.3%	+11.2%	11.8%	0.96 [0.56, 1.45]	1.54	-19%	64%	5.47	0.66	+1.8% (1.3)	7.0%
High-beta stocks (gross)	+18.8%	+17.7%	21.8%	0.86 [0.40, 1.40]	1.36	-34%	63%	3.82	1.36	-0.8% (-0.4)	12.5%
EW universe	+14.4%	+14.1%	15.1%	0.95 [0.52, 1.53]	1.52	-25%	68%	4.79	1.00	—	8.9%
Size-weighted universe	+15.4%	+15.3%	14.4%	1.07 [0.61, 1.64]	1.79	-24%	69%	5.27	0.90	+2.4% (1.8)	8.3%

EU (11 national blue-chip indices, point-in-time), median 238 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+4.4%	+3.6%	12.9%	0.34 [-0.12, 0.83]	0.42	-31%	55%	1.55	0.07	+3.6% (1.2)	7.8%
Long-only (net)	+9.5%	+9.4%	10.3%	0.93 [0.45, 1.50]	1.46	-19%	63%	4.20	0.62	+2.5% (2.2)	6.3%
Low-beta stocks (gross)	+9.7%	+9.6%	10.3%	0.95 [0.47, 1.52]	1.50	-19%	63%	4.29	0.62	+2.6% (2.3)	6.3%
High-beta stocks (gross)	+13.0%	+11.2%	21.6%	0.60 [0.15, 1.07]	0.85	-36%	58%	2.50	1.41	-3.1% (-2.1)	11.8%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
EW universe	+11.4%	+10.9%	14.5%	0.78 [0.33, 1.33]	1.22	-26%	64%	3.34	1.00	–	8.3%
Size-weighted universe	+10.7%	+10.1%	14.3%	0.75 [0.28, 1.28]	1.14	-26%	60%	3.12	0.95	-0.2% (-0.1)	8.3%

UK (FTSE 350, point-in-time), median 239 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+9.5%	+9.2%	11.8%	0.81 [0.29, 1.36]	1.27	-23%	63%	3.16	0.16	+7.9% (2.7)	7.4%
Long-only (net)	+10.4%	+10.4%	9.7%	1.07 [0.53, 1.76]	1.68	-19%	66%	4.81	0.63	+3.8% (4.1)	6.2%
Low-beta stocks (gross)	+10.6%	+10.6%	9.7%	1.09 [0.55, 1.78]	1.73	-19%	66%	4.92	0.63	+4.0% (4.3)	6.2%
High-beta stocks (gross)	+10.2%	+8.2%	21.0%	0.48 [0.02, 1.03]	0.60	-41%	56%	1.97	1.42	-4.7% (-3.2)	12.3%
EW universe	+10.5%	+9.9%	14.1%	0.74 [0.26, 1.34]	1.08	-29%	60%	3.23	1.00	–	8.8%
Size-weighted universe	+9.0%	+8.5%	12.7%	0.71 [0.21, 1.31]	1.01	-27%	62%	2.94	0.82	+0.4% (0.2)	7.7%

DK (OMXC25, point-in-time), median 19 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.9%	+1.4%	17.1%	0.17 [-0.29, 0.64]	0.12	-34%	55%	0.66	0.02	+2.6% (0.6)	10.7%
Long-only (net)	+11.4%	+11.1%	12.5%	0.91 [0.42, 1.44]	1.49	-18%	61%	3.61	0.64	+3.2% (1.6)	6.8%
Low-beta stocks (gross)	+11.6%	+11.4%	12.5%	0.93 [0.44, 1.46]	1.53	-17%	61%	3.68	0.64	+3.4% (1.7)	6.7%
High-beta stocks (gross)	+16.0%	+14.3%	22.6%	0.71 [0.17, 1.34]	1.04	-49%	60%	2.60	1.32	-1.0% (-0.4)	13.4%
EW universe	+12.9%	+12.3%	15.6%	0.82 [0.30, 1.44]	1.26	-33%	64%	3.21	1.00	–	9.3%
Size-weighted universe	+11.2%	+10.3%	16.6%	0.68 [0.11, 1.32]	0.94	-40%	61%	2.52	0.91	-0.6% (-0.2)	10.8%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 62 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.5%	+1.8%	12.2%	0.21 [-0.28, 0.68]	0.22	-28%	53%	0.87	0.05	+1.9% (0.6)	6.8%
Long-only (net)	+11.0%	+10.9%	10.8%	1.02 [0.49, 1.61]	1.71	-20%	64%	4.18	0.67	+2.4% (2.1)	6.0%
Low-beta stocks (gross)	+11.2%	+11.2%	10.8%	1.04 [0.51, 1.63]	1.75	-20%	64%	4.25	0.67	+2.6% (2.2)	6.0%
High-beta stocks (gross)	+14.7%	+13.5%	20.2%	0.73 [0.21, 1.36]	1.06	-43%	63%	2.84	1.35	-2.7% (-1.9)	12.0%
EW universe	+12.8%	+12.5%	14.1%	0.91 [0.39, 1.54]	1.42	-29%	67%	3.62	1.00	–	8.3%
Size-weighted universe	+11.2%	+10.7%	13.9%	0.80 [0.31, 1.37]	1.21	-27%	63%	3.26	0.95	-1.0% (-0.9)	8.6%

World (US + UK + EU, point-in-time), median 971 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+5.1%	+4.3%	13.2%	0.39 [-0.04, 0.87]	0.48	-24%	56%	1.84	0.26	+1.9% (0.6)	8.5%
Long-only (net)	+10.4%	+10.2%	11.6%	0.90 [0.46, 1.45]	1.37	-21%	67%	4.57	0.67	+1.9% (1.9)	7.1%
Low-beta stocks (gross)	+10.6%	+10.4%	11.6%	0.92 [0.47, 1.47]	1.41	-21%	67%	4.67	0.67	+2.1% (2.1)	7.1%
High-beta stocks (gross)	+14.8%	+13.1%	21.8%	0.68 [0.21, 1.23]	0.96	-38%	60%	2.78	1.34	-2.1% (-1.5)	12.7%
EW universe	+12.5%	+11.9%	15.7%	0.80 [0.33, 1.38]	1.20	-29%	64%	3.53	1.00	–	9.4%
Size-weighted universe	+11.4%	+10.8%	15.0%	0.76 [0.29, 1.33]	1.14	-28%	63%	3.23	0.94	-0.4% (-0.5)	9.0%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+11.4%	+6.7%	+39.8%	-12.4%	+4.7%	+21.4%

Year	US	EU	UK	DK	SCANDI	World
2014	+13.6%	+15.9%	+7.9%	-5.9%	+15.9%	+11.3%
2015	+6.7%	+12.5%	+30.0%	-7.0%	+7.1%	+12.3%
2016	-0.7%	+5.0%	+5.0%	-12.8%	-21.1%	-6.3%
2017	+7.8%	+13.3%	+22.4%	+26.3%	+10.7%	+18.6%
2018	-1.3%	+5.0%	-1.1%	-19.4%	+0.6%	-5.5%
2019	+16.1%	+11.8%	+18.6%	+58.9%	+18.0%	+20.8%
2020	-15.8%	-6.9%	+0.5%	-19.8%	-9.4%	-1.9%
2021	+16.8%	+5.3%	+15.3%	+31.8%	+10.9%	+14.7%
2022	+3.0%	-18.1%	-9.8%	+4.6%	-5.5%	-10.7%
2023	-24.1%	-6.7%	-13.1%	-12.9%	-17.4%	-13.0%
2024	+2.3%	+2.3%	+11.0%	-1.6%	+2.5%	+8.9%
2025	-14.1%	+2.9%	+5.5%	+8.2%	+6.0%	-4.5%
2026	-3.3%	+5.8%	+5.1%	+9.7%	+11.4%	+1.4%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+22.2%	+14.4%	+25.7%	+18.1%	+16.7%	+24.6%
2014	+17.5%	+10.3%	+6.6%	+8.3%	+16.2%	+9.6%
2015	+2.4%	+11.8%	+16.8%	+13.7%	+11.5%	+5.0%
2016	+9.7%	+11.7%	+14.9%	+3.4%	+11.6%	+3.1%
2017	+14.6%	+14.8%	+21.3%	+25.8%	+21.5%	+20.4%
2018	-2.4%	-3.5%	-3.8%	-14.2%	-3.6%	-6.8%
2019	+27.9%	+23.1%	+22.8%	+41.7%	+31.2%	+26.0%
2020	+10.5%	+3.5%	+1.6%	+15.2%	+10.4%	+12.3%
2021	+23.9%	+14.7%	+14.6%	+27.8%	+19.9%	+18.1%
2022	-3.7%	-11.6%	-11.0%	-7.5%	-12.5%	-10.7%
2023	+2.3%	+9.7%	+2.4%	+4.3%	+0.9%	+7.3%
2024	+13.9%	+6.5%	+10.5%	+3.1%	+4.2%	+11.7%
2025	+6.1%	+16.0%	+14.0%	+14.8%	+16.0%	+13.7%
2026	+8.9%	+10.8%	+11.3%	+8.7%	+11.9%	+10.6%

7. Trading record

The rebalance log per universe is in `results/xs_FS10_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	249 / 249	13%	6%	52%	-12.5%	2021-02
EU	163	117 / 117	14%	5%	55%	-11.6%	2021-09
UK	163	119 / 119	16%	6%	63%	-11.2%	2021-02
DK	163	9 / 9	15%	6%	55%	-13.6%	2022-05
SCANDI	163	31 / 31	13%	6%	53%	-7.8%	2021-02
World	163	483 / 483	16%	6%	56%	-11.0%	2021-02

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: low-beta stocks	Short: high-beta stocks
US	CF, OXY, XOM, EOG, APA, FANG, CVX, DVN	MRNA, SMCI, SNDK, CCL, COIN, HOOD, UAL, Q
EU	REP.MC, NESTE.HE, GALP.LS, ENI.MI, TTE.PA, SHELL.AS, JSW.WA, KPN.AS	MTS.MC, MT.AS, IFX.DE, KGH.WA, ENR.DE, IAG.MC, BOL.ST, GLE.PA
UK	HBR.L, ITH.L, BP.L, SHELL.L, ENOG.L, BEZ.L, BHMGL, IMB.L	RPI.L, GDWN.L, RHIM.L, WIZZ.L, ANTO.L, FRES.L, HOC.L, PAF.L
DK	TRYG.CO, NSIS-B.CO, JYSK.CO, ISS.CO, NDA-DK.CO, CARL-B.CO, DANSKE.CO, RBREW.CO	ZEAL.CO, GN.CO, NOVO-B.CO, AMBU-B.CO, ROCK-B.CO, ORSTED.CO, DEMANT.CO, DSV.CO
SCANDI	NESTE.HE, ELISA.HE, TELIA.ST, TRYG.CO, TEL2-B.ST, EVO.ST, MAERSK-B.CO, ESSITY-B.ST	KCR.HE, METSO.HE, BOL.ST, ROCK-B.CO, GN.CO, ZEAL.CO, QTCOM.HE, SAND.ST

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | ---|---|---|---|---|---| | US | JKP USA 7 themes | -6.1% | -2.86 | 0.63 | mkt +0.85 (+9.0), value -0.44 (-2.9), low_risk +1.70 (+8.8) | | EU | French Europe 5F + WML | -2.3% | -0.73 | 0.29 | Mkt-RF +0.24 (+4.0), RMW +1.15 (+4.1), CMA +0.56 (+2.4), WML +0.32 (+2.7) | | UK | JKP GBR 7 themes | +3.9% | 1.28 | 0.48 | mkt +0.39 (+7.0), momentum +0.26 (+2.7), low_risk +0.93 (+5.5), quality +0.64 (+4.0) | | DK | JKP DNK 7 themes | +0.1% | 0.03 | 0.40 | mkt +0.43 (+5.9), size -0.33 (-2.2), low_risk +1.22 (+8.5) | | SCANDI | French Europe 5F + WML | -3.1% | -1.03 | 0.12 | Mkt-RF +0.13 (+2.4), RMW +0.57 (+2.3), WML +0.33 (+3.5) | | World | JKP World 7 themes | +0.8% | 0.32 | 0.45 | mkt +0.67 (+7.8), momentum +0.31 (+2.4), low_risk +1.24 (+7.7) |

Long-only

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | |---|---|---|---|---|---| | US | JKP USA 7 themes | +1.7% | 1.73 | 0.88 | mkt +0.91 (+31.5), size -0.24 (-3.5), value -0.14 (-2.5), low_risk +0.71 (+8.8) | | EU | French Europe 5F + WML | +4.0% | 2.09 | 0.65 | Mkt-RF +0.54 (+15.5), RMW +0.44 (+3.1) | | UK | JKP GBR 7 themes | +5.0% | 2.67 | 0.68 | mkt +0.46 (+12.3), quality +0.40 (+4.0) | | DK | JKP DNK 7 themes | +5.9% | 2.45 | 0.58 | mkt +0.62 (+12.6), value +0.10 (+2.0), low_risk +0.25 (+3.1), quality +0.15 (+2.4) | | SCANDI | French Europe 5F + WML | +6.2% | 2.72 | 0.51 | Mkt-RF +0.50 (+10.2), RMW +0.46 (+3.6) | | World | JKP World 7 themes | +5.0% | 3.07 | 0.77 | mkt +0.70 (+15.3), low_risk +0.49 (+6.7) |

9. Statistical verdict

- **Gate:** |t| > 2.87 (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in all six; passes in 1 of 6 (UK).
- **L/S CAPM alpha:** -0.8% to +7.9%, |t| past the gate in 0 of 6.
- **Long-only:** Sharpe above the equal-weight universe in 5 of the six; alpha +1.6% to +3.8%, passing in 1 of 6.
- **Probabilistic Sharpe** of the L/S, P(true Sharpe > 0): US 0.67, EU 0.89, UK 1.00, DK 0.73, SCANDI 0.78, World 0.92.
- **Publication decay:** gross L/S +4.2% to +18.6% a year in 2013–19 against +0.7% to +8.1% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	+10.2%	0.01	+0.1%	+10.1%	-0.5%	-1.3%	-0.7%	19%	2013-05 -7%, 2018-02 -6%, 2016-08 -6%
EU	+12.2%	-0.03	-0.4%	+12.6%	-0.4%	+0.1%	+0.4%	18%	2016-10 -6%, 2019-04 -5%, 2017-06 -4%
UK	+18.6%	0.38	+5.0%	+13.6%	-0.6%	+1.1%	-0.3%	23%	2016-11 -6%, 2016-06 -5%, 2018-12 -4%
DK	+4.2%	0.19	+2.9%	+1.4%	-0.5%	-2.2%	-5.6%	21%	2016-10 -11%, 2018-10 -11%, 2013-05 -11%

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
SCANDI	+7.1%	-0.00	-0.0%	+7.1%	-0.5%	-1.0%	-3.3%	20%	2016-11 -7%, 2016-10 -7%, 2013-08 -6%
World	+12.3%	0.13	+1.6%	+10.7%	-0.6%	-1.0%	-0.1%	23%	2018-05 -6%, 2018-02 -5%, 2016-08 -5%

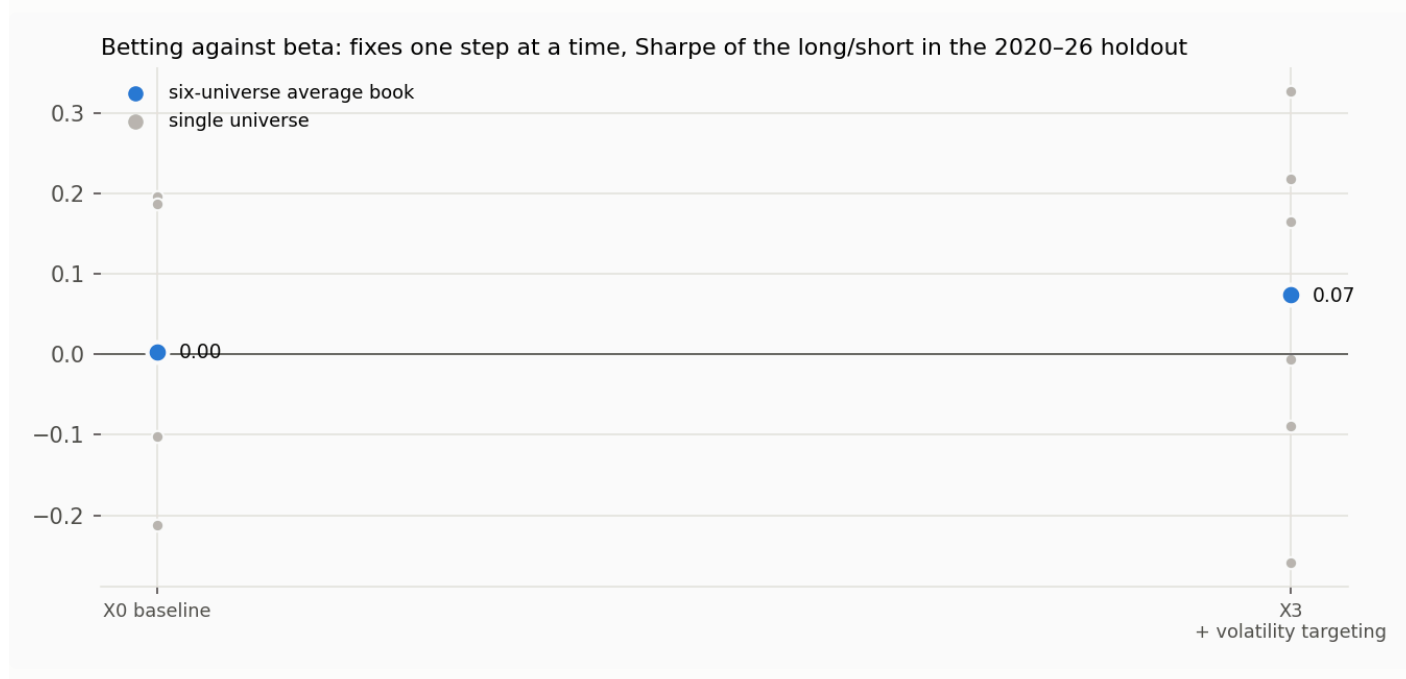
On average across the six universes the gross spread was +10.8% a year, of which the market exposure (beta +0.11) contributed +1.5%; the beta-adjusted spread (CAPM alpha) was +9.2%. Costs took 0.5% a year at 21% monthly turnover across both legs. The long leg beat the universe by -0.7% and the short leg lagged it by -1.6% a year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +5.2%, +5.6%, +6.0% a year.

10.2 The fix ladder

X0 baseline → X3 volatility targeting (BAB is beta-neutral and rank-weighted by construction, so the beta-neutral and buffer steps do not apply).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	0.81	—	0.00	—	—	+0.0%	-22%	-1.5% (-0.4)	0.5%
X3 + volatility targeting	0.70	+0.03 (p 0.467)	0.07	+0.07 (p 0.329)	4 of 6	+0.8%	-22%	-0.4% (-0.2)	0.5%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.

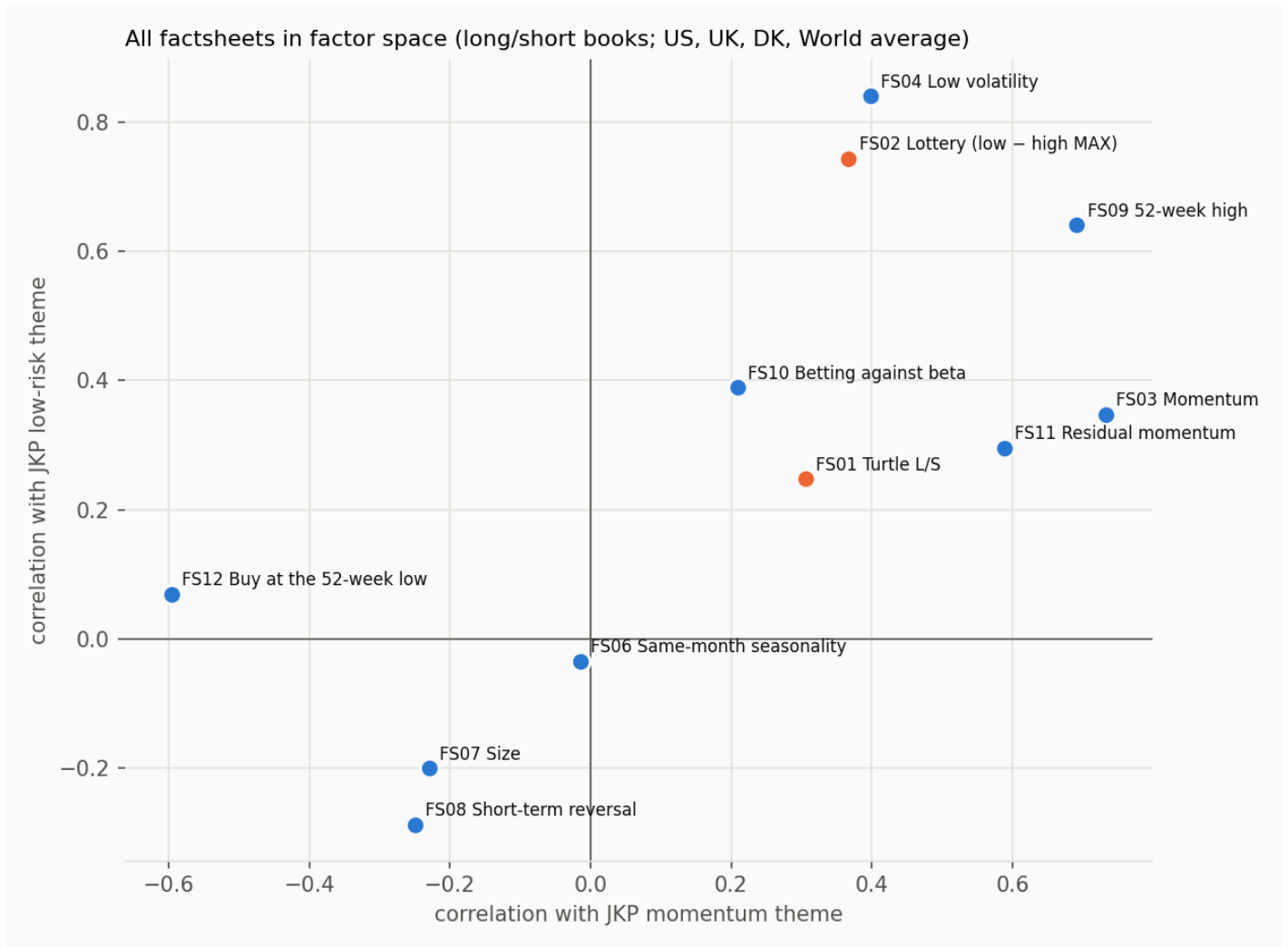


- **X3 + volatility targeting:** +0.03 design, +0.07 holdout; helps in both windows but does not pass the gate.

Where it ends: holdout Sharpe 0.00 → 0.07, return +0.0% → +0.8% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Betting against beta
Series family	Low risk
Academic style	Defensive / betting against beta
Signal	Price only (daily total returns)
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 21% of the two legs replaced monthly
Market exposure	L/S beta 0.02 to 0.26
Payoff shape	Treynor–Mazuy γ -2.25 to -0.59 (t -5.1 to -0.7); worst 10% of market months -3.2% to -0.2% a month, best 10% -0.8% to +0.8%
Nearest factor theme	low_risk (2 of 6 universes), RMW (1 of 6 universes)
Nearest library signals (returns)	Volatility (252 days) (6 universes), Volatility (6 universes), Range (MAX–MIN) (5 universes)
Economic rationale	Leverage-constrained investors buy high-beta stocks instead of leveraging the market, bidding them up (Black 1972; Frazzini & Pedersen 2014); lottery demand (Bali, Brown, Murray & Tang 2017).
Publication	Black, Jensen & Scholes 1972; Frazzini & Pedersen 2014; critique by Novy-Marx & Velikov (2022) on construction and costs
Where it fits	In the low risk cluster of the library; with some information beyond its neighbours.

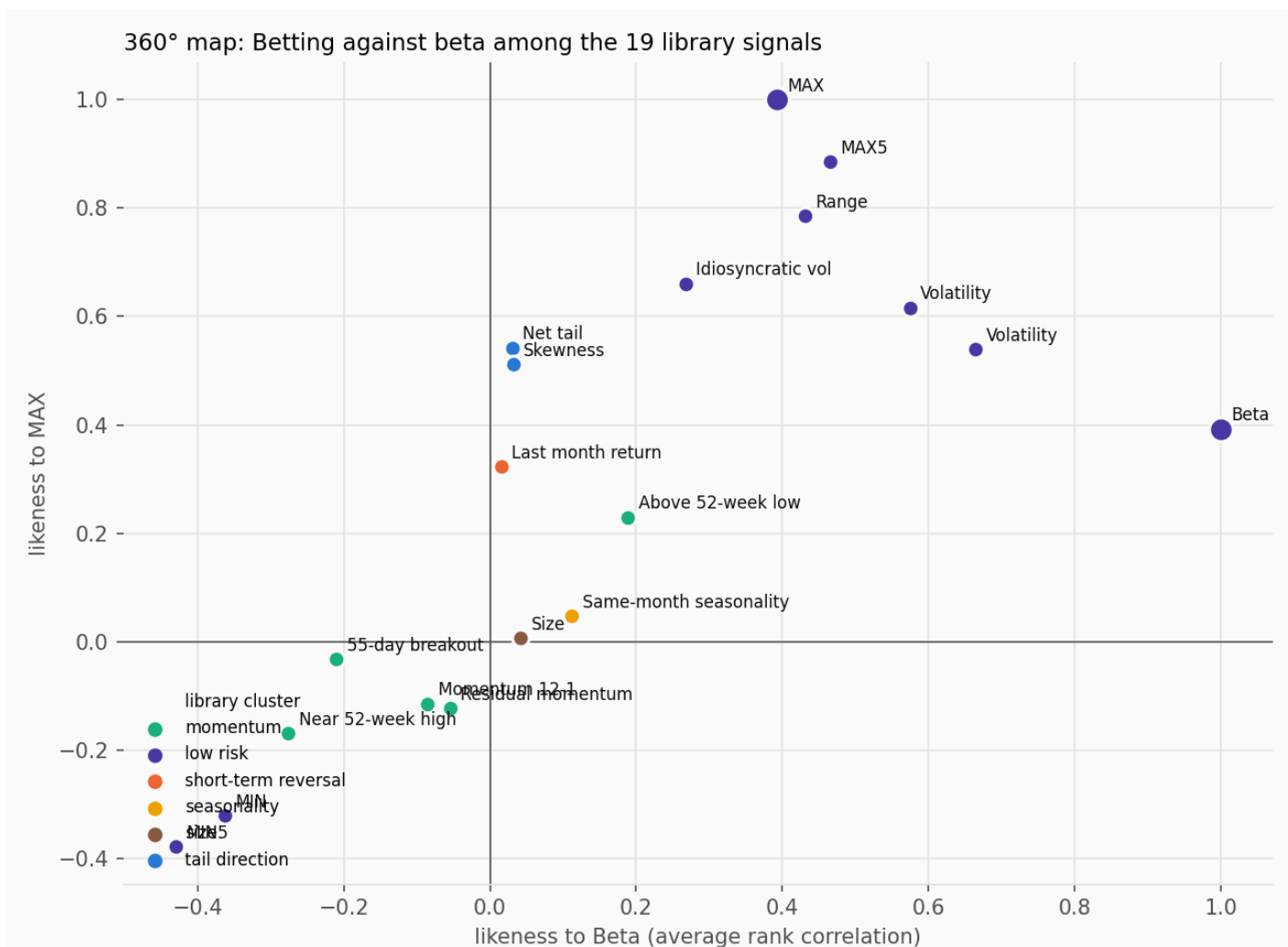


12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **MAX**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha t range	Universes alpha t $\geq +2$ / ≤ -2
Beta (this strategy)	low risk	+1.00	+1.00	+5.5%	-15.7% to -2.6%	0 / 3
Volatility (252 days)	low risk	+0.66	+0.85	+5.1%	-10.3% to -0.2%	0 / 1
Volatility	low risk	+0.58	+0.83	+3.0%	-10.3% to -3.7%	0 / 5
Range (MAX–MIN)	low risk	+0.43	+0.75	+1.2%	-11.3% to -2.3%	0 / 4
MIN5	low risk	-0.43	-0.73	-2.0%	+2.4% to +13.2%	4 / 0
MAX5	low risk	+0.47	+0.71	+2.6%	-8.5% to +0.4%	0 / 2

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
MAX (best day) (mirror)	low risk	+0.39	+0.70	+1.9%	-9.3% to +1.0%	0 / 2
MIN (worst day)	low risk	-0.36	-0.68	-1.5%	+2.9% to +11.0%	3 / 0
Near 52-week high	momentum	-0.28	-0.65	+2.5%	+8.3% to +18.8%	6 / 0
Idiosyncratic vol	low risk	+0.27	+0.59	-1.5%	-10.4% to -0.8%	0 / 3
55-day breakout	momentum	-0.21	-0.56	+0.7%	+5.6% to +12.6%	3 / 0
Momentum 12-1	momentum	-0.09	-0.39	+7.7%	+3.9% to +17.8%	5 / 0
Residual momentum	momentum	-0.05	-0.25	+6.1%	+7.3% to +13.0%	5 / 0
Last month return	short-term reversal	+0.02	-0.22	+0.9%	-3.2% to +8.4%	0 / 0
Size	size	+0.04	-0.21	-1.5%	-3.8% to +4.6%	0 / 0
Skewness	tail direction	+0.03	+0.10	-0.2%	-5.4% to +4.5%	0 / 1
Same-month seasonality	seasonality	+0.11	-0.04	-1.8%	-8.5% to +7.2%	0 / 0
Above 52-week low	momentum	+0.19	+0.04	+8.8%	+4.4% to +13.8%	4 / 0
Net tail (MAX+MIN)	tail direction	+0.03	+0.01	-1.0%	-5.3% to +4.2%	0 / 0



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

| Universe | Alpha after neighbours / yr | t | R² | Neighbours with |t| ≥ 2 | ---|---|---|---|---| | US | -1.7% | -0.68 | 0.86 | Volatility +0.93 (+7.1) | | EU | -4.3% | -1.20 | 0.72 | Volatility +0.47 (+2.5), Volatility +0.37 (+2.0) | | UK | -9.6% | -3.62 | 0.83 | Volatility +0.59 (+4.9) | | DK | +3.9% | 1.27 | 0.76 | Volatility +0.69 (+8.3), Volatility +0.17 (+2.2) | | SCANDI | -0.6% | -0.24 | 0.64 | Volatility +0.38 (+2.4) | | World | -2.5% | -1.16 | 0.86 | Volatility +0.87 (+7.1) |

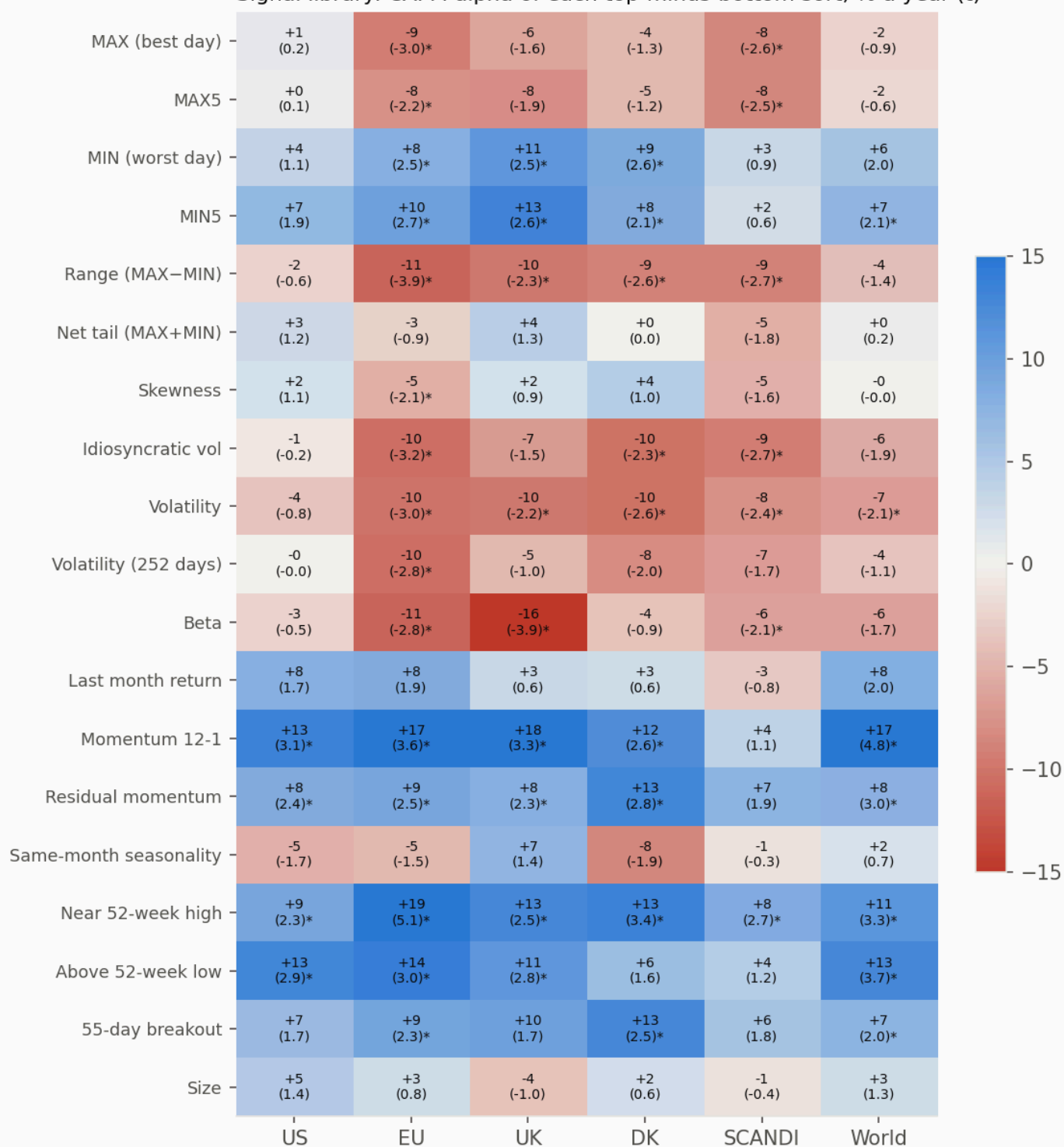
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Beta	10.8%	10.5%	11.3%
middle	12.1%	12.7%	10.7%
high Beta	14.4%	14.4%	13.8%
	low MAX	middle	high MAX

Holding MAX fixed (down a column), moving from low to high Beta changes the return by +3.6, +3.9 and +2.6 points a year; holding Beta fixed (along a row), moving from low to high MAX changes it by +0.5, -1.5 and -0.5. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): +32% vs -42% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +5.5%, +5.2%, +5.6%, +6.0% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.

4. **Data gap.** Financing uses the USD risk-free rate for all markets, which overstates the cost in EU and Denmark while their rates were negative (2015–21). Betas are raw, not shrunk.
5. **Not a registered trial.** Descriptive; the fix ladder is counted inside FS03–FS12 (25 trials).

14. Academic references

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15. Reproduce

`code/data.py` → `signals.py`, `signals2.py`, `library.py` → `xs.py` + `analyse_xs.py` FS10 → `build_xs.py`
FS10. Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.